

Programming Using C++ Level 1

For Loops



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Definition

- ◆ A for loop is a repetition statement used to execute a block of code multiple times.
- ◆ A for loop is useful when the number of repetitions is known or controlled by a counter.
- ◆ In this program, the loop prints numbers from 1 to N.

Explanation

- ◆ The program asks the user to enter a number N .
- ◆ The for loop starts with $i = 1$ and continues while $i \leq N$.
- ◆ After each iteration, $i++$ increases the value of i by 1.
- ◆ The loop stops when i becomes greater than N .

Steps in main()

1 Define the function

1 Read the input

◆ Read the user's number and store it in N.

2 Initialize the counter

◆ Start the loop with $i = 1$.

3 Check the condition

◆ Continue while $i \leq N$.

4 Execute the loop body

◆ Print the current value of i .

5 Increment the counter

◆ Use $i++$ to increase i by 1.

6 Stop the loop

◆ When i becomes greater than N , the loop ends.

💡 Example

◆ If the user enters:

N = 5

◆ The loop produces:

1

2

3

4

5

➡ Result

◆ The program prints all numbers starting from 1 up to the number entered by the user.

◆ For N = 5:

1 2 3 4 5

Note

- ◆ The general structure of a for loop is:
for (initialization; condition; update)
- ◆ In this program:
for (int i = 1; i <= N; i++)
- ◆ int i = 1 initializes the counter.
- ◆ i <= N is the condition.
- ◆ i++ increases the counter after each iteration.

⊘ Warning

- ◆ Make sure the loop condition can eventually become false.
- ◆ If the condition always remains true, the loop can become an infinite loop.



Comparison

for Loop

- ✓ Commonly used when a counter controls the repetitions.
- ✓ Initialization, condition, and increment are written together.
- ✓ Makes counting loops easy to read.

Without a Loop

- ✓ Each number would need to be printed separately.
- ✓ More code would be required.
- ✓ Repeating or changing the range would be harder.

✦ Summary

- ✓ A for loop repeats a block of code while its condition is true.
- ✓ A counter can control how many times the loop runs.
- ✓ A for loop commonly contains initialization, condition, and update.

Thank you 😊